CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

"*"Barcode Area"*"
Front Page Bar Code

18 JANUARY 2021 (a.m.)

FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE | 0 | 1 | 2 | 2 | 9 | 0 | 2 | 0 |

SUBJECT INFORMATION TECHNOLOGY – Paper 02

PROFICIENCY GENERAL

REGISTRATION NUMBER | | | | | | | | | | |

SCHOOL/CENTRE NUMBER

NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

"*"Barcode Area"*"
Current Bar Code

DATE OF BIRTH | D | D | M | M | Y | Y | Y | Y |

SIGNATURE _____

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE ON THIS PAGE

FORM TP 2021016

TEST CODE **01229020**

JANUARY 2021

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2020 Caribbean Examinations Council
All rights reserved.

"*"Barcode Area"*"
Sequential Bar Code

Answer ALL questions.

1. The Caribbean Development Bank plans to fund a project to establish a meteorological station in a Caribbean island, to forecast and monitor weather conditions in the region. The type of computer system suitable for this project is a supercomputer.

(a) State TWO reasons why a supercomputer is suitable for this project.

.....

.....

.....

.....

(2 marks)

(b) The local area networks of stakeholders throughout the Caribbean region are to be connected to the supercomputer to form a wide area network, using telecommunication lines.

(i) Identify the MOST important hardware device that is necessary for this connection.

.....

.....

(ii) State the role of the hardware device identified in (b) (i).

.....

.....

.....

(2 marks)

(c) State TWO examples of industries, other than weather forecasting, in which supercomputers are still being used.

.....

.....

(2 marks)

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(d) State TWO characteristics of desktop or personal computers that make them popular among computer users.

.....

.....

.....

.....

.....

(2 marks)

(e) State TWO areas where desktop or personal computers are widely used.

.....

.....

(2 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

"*"Barcode Area"*"
Sequential Bar Code

(f) The information in Worksheet 1 below shows the total sales for each of the months January, February and March, for five salespersons in their particular districts.

For example, Kathleen Albert, a salesperson from the Port-of-Spain district has sales of $1925, $1500 and $1350 for the months of January, February and March respectively.

The data in range A1:E8 of Worksheet 2 was summarized and reorganized from Worksheet 1, the original worksheet, to allow for easier reading.

Worksheet 1

	A	B	C	D
1	**Salesperson**	**District**	**Sales**	**Month**
2	Albert, Kathleen	Port-of-Spain	$1 925.00	January
3	Albert, Kathleen	Port-of-Spain	$1 500.00	February
4	Albert, Kathleen	Port-of-Spain	$1 350.00	March
5	Jordan, Michael	Diego Martin	$1 400.00	January
6	Jordan, Michael	Diego Martin	$1 550.00	February
7	Jordan, Michael	Diego Martin	$1 400.00	March
8	Clement, Davis	Marabella	$1 235.00	February
9	Clement, Davis	Marabella	$1 850.00	January
10	Clement, Davis	Marabella	$1 600.00	March
11	Charles, Johnson	Siparia	$1 400.00	January
12	Charles, Johnson	Siparia	$1 965.00	February
13	Charles, Johnson	Siparia	$1 125.00	March
14	Weekes, Terecita	Long Circular	$2 500.00	March
15	Weekes, Terecita	Long Circular	$1 305.00	January
16	Weekes, Terecita	Long Circular	$1 435.00	February

Worksheet 2

	A	B	C	D	E
1	**Sum of Sales**	**Column Labels**			
2	**Row Labels**	**January**	**February**	**March**	**Grand Total**
3	Albert, Kathleen	1925	1500	1350	4775
4	Charles, Johnson	1400	1965	1125	4490
5	Clement, Davis	1850	1235	1600	4685
6	Jordan, Michael	1400	1550	1400	4350
7	Weekes, Terecita	1305	1435	2500	5240
8	**Grand Total**	**7880**	**7685**	**7975**	**23540**
9					
10	Clement, Davis				

GO ON TO THE NEXT PAGE

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(i) Identify the feature/function of the spreadsheet which was used to produce the summarized data in Worksheet 2.

.....

(1 mark)

(ii) State the information that is provided when a value in the Grand Total row of Worksheet 2 is double clicked.

.....

.....

(1 mark)

(iii) If the values in Worksheet 1 are updated, the values in Worksheet 2 are not automatically updated. State the TWO steps required, in the correct order, to manually update the values in Worksheet 2.

.....

.....

.....

(2 marks)

(iv) Complete the missing parameters in the VLOOKUP formula for Cell B10 of Worksheet 2, to find the amount of sales made during the month of March by the salesperson in Cell A10.

= VLOOKUP (_____, _____, _____, _____).

(5 marks)

(v) State the change that should be made to Parameter 3 in (f) (iv), in order to obtain the Grand Total for the salesperson in Cell A10.

.....

(1 mark)

Total 20 marks

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Jay is the information and communications technology (ICT) manager at Hemprezz Power, a renewable energy company. The proprietor has announced that the company will be expanding, with new branches opening in the neighbouring city.

(a) Jay has outlined to the proprietor that in order to facilitate the expansion multiple computer systems would be needed.

(i) List the FOUR basic components that each computer system should have.

.....

.....

(4 marks)

(ii) Jay has listed a number of specialized peripheral devices that will be required. Complete the table below by indicating the category of EACH device.

Device	Category
Joystick	
Scanner	

(2 marks)

(b) Jay will need to set up a wired network to facilitate communication between the computer systems.

(i) List THREE types of transmission media that could be used to facilitate the communication.

.....

.....

(3 marks)

(ii) Identify the transmission media in (b) (i) which would be MOST appropriate for use in the scenario described.

.....

.....

(1 mark)

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) In presenting his proposal for the expansion to the proprietor, Jay prepares the following spreadsheet which outlines the costs of materials for one of the new branches, Rock Site Startup.

	A	B	C	D	E
1	Rock Site Startup				
2	Items	Item Cost	Number	Total Cost	
3					
4	Tables (Type 1)	5 000	12		
5	Tables (Type 2)	18 000	2		
6	Chairs (Type 1)	2 000	24		
7	Chairs (Type 2)	2 500	15		
8	Projector	25 000	2		
9	White board	30 000	2		
10	Smart screen	50 000	1		
11	Lighting	1 200	15		
12	Computer systems	38 000	15		
13					
14	Total cost				
15					
16					

Write the formula which is required to calculate the total cost in EACH of the following cells.

(i) Cell D4

.....

.....

(2 marks)

(ii) Cell D14

.....

.....

(2 marks)

GO ON TO THE NEXT PAGE

"*"Barcode Area"*"
Sequential Bar Code

(d) To improve the presentation of the document, Jay formatted Cell A1 across all columns as shown in the spreadsheet in (c) on **page 9**.

(i) Identify the formatting feature used.

.....

.....

(1 mark)

(ii) List the steps taken by Jay, in the correct order, to apply the formatting feature identified in (d) (i).

.....

.....

.....

(3 marks)

(e) State the steps, in the correct order, required to create a formula to find the average of a range of numbers in a spreadsheet.

.....

.....

.....

(2 marks)

Total 20 marks

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Chavvi was nominated to serve as the team lead for planning an ICT symposium, hosted by Hemprezz Power.

(a) List THREE groups of ICT- related personnel that Chavvi will need to invite to the symposium.

.....

.....

.....

(3 marks)

(b) State TWO ways in which the extensive use of ICT has positively impacted the field of business.

.....

.....

.....

(2 marks)

Chavvi was asked to prepare a memorandum to the director general outlining the plans for the symposium.

The following is an extract from the memorandum prepared using the Word format.

Memorandum

TO: Mrs Judy Staton
Director General

FROM: Chavvi Sampson
Head, Cybersecurity

DATE: 2 April 2020

SUBJECT: Cybersecurity Exercise 2020

In an increasingly connected world, it is necessary to defend the critical infrastructure of each country, to understand how the attacker works, to know what to look for within a data centre infrastructure and to have the ability to address the business security strategy.

For approximately eight hours, the multidisciplinary teams will compete with each other to achieve the full range of challenges and defence, with an expert's guidance.

Participant Profile

It is recommended that the participants have responsibility for the following

- Networks
- Network communications
- Infrastructure

It is recommended that they have prior knowledge in the following:

- Basic Windows and Linux commands
- Networks knowledge
- Basic threat knowledge
- Basic firewall and IPS configuration
- Security
- Incident response

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(c) State ONE reason for using bullets and numbering in a Word document.

.....

.....

(1 mark)

(d) Other than bullets and numbering, identify TWO formatting features used in the extract from the memorandum.

.....

.....

.....

(2 marks)

(e) Describe how the positions of paragraphs 1 and 2 from the extract can be interchanged, using the drag and drop feature.

.....

.....

.....

.....

.....

(3 marks)

(f) Describe the steps used in conducting a spell check in a Word document.

.....

.....

.....

.....

.....

.....

.....

.....

(4 marks)

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(g) The pseudocode below will calculate and output the number of participants and the average age of all participants attending the ICT symposium. The algorithm will terminate and display the output when an age of 999 is entered.

Complete the following pseudocode by filling in the blank spaces, labelled 1–10, with the MOST appropriate variables and/or reserve words.

```
Start
        Set COUNT ← 0
        Set AGE ← 0
        Set AVERAGE ← 0
        Set TOTAL ← 0

_____________________________ "Enter Participant's age"
                1

Read AGE
        WHILE AGE <> _____________________   _____________________
                              2                        3

                Set TOTAL ← TOTAL + AGE

                Set COUNT ← COUNT + _____________________________________
                                                   4

                Set _____________________ ← TOTAL / COUNT
                              5

                Write "Enter another age or 999 to terminate the program"

                Read _____________________
                              6

                _____________________
                              7

                Write "The Number of Participants are: " , _________________
                                                            8

                Write "The average age of Participants are: " , _________________
                                                                  9

                _____________________
                              10
```

(10 marks)

Total 25 marks

GO ON TO THE NEXT PAGE

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. (a) The Stennet Marketing Company collects personal data from customers which it stores on computers.

(i) State TWO possible effects of the improper storage of data.

.....

.....

.....

.....

.....

(2 marks)

(ii) State TWO measures that can be used to protect the data stored on a computer.

.....

.....

.....

.....

.....

(2 marks)

"*"Barcode Area"*"
Sequential Bar Code

(b) (i) Outline the function of the loop construct in computer programming.

.....

.....

.....

.....

(2 marks)

(ii) State TWO distinguishing features for EACH of the following types of loops.

- WHILE . . . ENDWHILE

.....

.....

.....

- REPEAT . . . UNTIL

.....

.....

.....

(4 marks)

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(c) You are hired as a programmer by Stennet Marketing Company, to write a pseudocode for a supermarket. This pseudocode would be presented to the company, before the source code is written. The requirements for the pseudocode are as follows:

- Prompt the cashier to enter the price and quantity of items.
- Calculate the discount amount at 5% of the total, if the total is greater than $500.
- Otherwise, calculate the discount amount at 10% of the total.
- The algorithm should output the total and discount.

Write the pseudocode with the requirements as outlined above.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(10 marks)

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(d) Write the pseudocode, using NESTED IF STATEMENTS, to do the following:

If the mark entered is greater than or equal to 70, then display the statement "GRADE A". Otherwise, if the mark entered is greater than or equal to 60, then display the statement "GRADE B". For all other marks entered, display the statement "GRADE NOT AVAILABLE".

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(5 marks)

Total 25 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

"*"Barcode Area"*"
Sequential Bar Code

EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

"*"Barcode Area"*"
Sequential Bar Code

EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

"*"Barcode Area"*"
Sequential Bar Code

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE ON THIS PAGE

CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE:

1. **Fill in all the information requested clearly in capital letters.**

TEST CODE: | 0 | 1 | 2 | 2 | 9 | 0 | 2 | 0 |

SUBJECT: INFORMATION TECHNOLOGY – Paper 02

PROFICIENCY: GENERAL

REGISTRATION NUMBER: | | | | | | | | | | |

FULL NAME: _____

(BLOCK LETTERS)

Signature: _____

Date: _____

2. **Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.**

3. **Keep it in a safe place until you have received your results.**

INSTRUCTION TO SUPERVISOR/INVIGILATOR:

Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

Signature: _____

Supervisor/Invigilator

Date: _____